

Day 5 Labs - SIEM Fundamentals



What We Learned So Far — Labs 1–9

- Labs 1–6: TI platforms, OSINT, Nmap, CVEs
- Lab 7: Phishing analysis, email authentication
- Labs 8–9: Password cracking, C2 detection, DNS tunnelling, exfiltration
- Today we practice: SIEM architecture, Windows/Linux log collection, correlation rules, alert triage

Lab 10 Overview

- **Security Information and Event Management**
- SIEM architecture concepts
- Log collection basics
- Simple correlation
- Alert investigation
- **Duration: 2 hours**

SIEM Architecture

- **Core components:**

- **Log sources:**

Log Sources → Collection → Parsing →
Storage → Correlation → Alerting → Dashboard

- Windows Event Logs

- Linux syslog

- Firewall logs

- Application logs

- Network devices

Windows Event Forwarding

- **Send Windows logs to SIEM:**

Key Event IDs:

4624 - Successful logon

4625 - Failed logon

4672 - Special privileges assigned

4688 - Process creation

4720 - User account created

7045 - Service installed

Linux Log Collection

- **Important Linux logs:**

/var/log/auth.log	- Authentication
/var/log/syslog	- System events
/var/log/secure	- Security events
/var/log/apache2/	- Web server

- **Forward via:**

- rsyslog
- Filebeat
- Syslog-ng

Exercise - Configure Log Forwarding

- **On Windows (DC01):**
 - 1. Enable audit policies (Lab 3)
 - 2. Configure Windows Event Forwarding
 - 3. Verify events generating
- **On Linux:**

```
# View authentication logs
sudo tail -f /var/log/auth.log

# Generate events
sudo su - testuser
```

Simple Correlation Rules

- **Detection logic examples:**

Rule: Brute Force Detection

Condition: >5 Event ID 4625 from same IP in 5 minutes

Action: Alert, priority HIGH

Rule: New Admin Account

Condition: Event ID 4720 followed by 4732
(Administrators)

Action: Alert, priority CRITICAL

Exercise - Create Detection Rules

- **Design rules for:**
 - 1. Multiple failed logins (brute force)
 - 2. Login outside business hours
 - 3. Service installation (persistence)
 - 4. Process execution from temp folder
- **Document: Condition, threshold, action**

Alert Investigation Process

- **When alert fires:**
 - 1. **Triage:** Is this true positive or false positive?
 - 2. **Scope:** What systems affected?
 - 3. **Context:** What was user doing?
 - 4. **Escalate or close:** Based on findings
- **Document everything**

Exercise - Alert Triage

- **Investigate sample alert:**

```
Alert: Multiple failed logins  
Source IP: 192.168.100.20  
Target: DC01  
User: Administrator  
Count: 15 in 2 minutes
```

- **Questions to answer:**
- 1. What is source system?
- 2. Is this expected behavior?
- 3. Did any logins succeed after failures?

Lab 10 Validation

- **Verify your SIEM knowledge:**
 - [] Understand SIEM architecture
 - [] Can identify key Windows Event IDs
 - [] Know Linux log locations
 - [] Designed correlation rules
 - [] Performed alert triage exercise
 - [] Documented investigation process

Lab 10 Summary — SIEM & Log Analysis

- Understood SIEM data flow: sources → collection → parsing → correlation → alert
- Identified critical Windows Event IDs for authentication and process monitoring
- Located and interpreted key Linux log files (/var/log/auth.log, syslog)
- Designed correlation rules for brute force, after-hours logins, persistence
- Completed alert triage exercise — distinguished true positives from false positives

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Thank You

